Modbus Protocol

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Revision History

**The version is denoted with 1.0, or 1.1 or 1.2...*

Version	Date	Description	Prepared	Approved
1.2.4	2025-03-10	1. Add string voltage and current	Devin Wang	Ted Chen
1.2.3	2025-01-16	1. Add 0x0835 2. Modify MPPT and load current form U16 to S16 3. Modify string current from U16 to S16	Devin Wang	Ted Chen
1.2.2	2024-12-2	1. Add 0x1211register address 2. Add 0x1A8E register address 3. Add 0x600D register address 4. Add 0x8000~8B3F register address 1. Add 0x1034~1035register address 5. Add 0x513D~513Eregister address 6. Add LCD all register address	Devin Wang	Ted Chen
1.2.1	2024-11-15	1. Add 0x03E8~03FD register address 2. Add 0x07D0~07D3, 0x0834 register address 3. Add modbus protocol command,Read Register (0x7E)	Devin Wang	Ted Chen
1.2	2023-11-8	1. Delete firmware build date 2. Modify PV1~PV4 to MPPT1~4 3. Add all parameter setting range value 4. Modify 0x2009/0x200A type from U32 to S32 5. Modify 0x2100 item 2 definition 6. Add 0x2106~210F, 2122~213C, 2141, 2143, 2146~2149 register address 7. Add 0x5021/5022/5029/5063~5067/506C/506D/507F/5081~5084 register address 8. Delete 0x502B, 5051~5058 register address 9. Add 0x5106/5107 register address 10. Delete 0x510F/5110/5111/5113 register address 11. Add 0x600F/6010 register address 12. Modify 0x2136 register address unit from hour to minute 13. Add 0x2150 register address 14. Add 0x2168~217F register address 15. Add 0x3005 register address 16. Add 0x5101 register address 0x48 for EN50549-1：Ireland	Devin Wang	Ted Chen

		17. Charger and Discharger word delete at last "r" 18. Add 0x30B1 setting value 9 for EM530 digital meter, 10 for EM540 digital meter 19. Add the note about the nominal voltage and nominal power setting range		
1.1	2023-3-20	1. Update error code 2. Add 0x1080~1092 MPPT5~9 register address	Devin Wang	Ted Chen
1.0	2021-1-2	Initial version	Devin Wang	Yanming Jia

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1 Protocol Description

The Modbus communication adopts 2-line RS-485 interface, and a single host can connect up to 247 inverters.

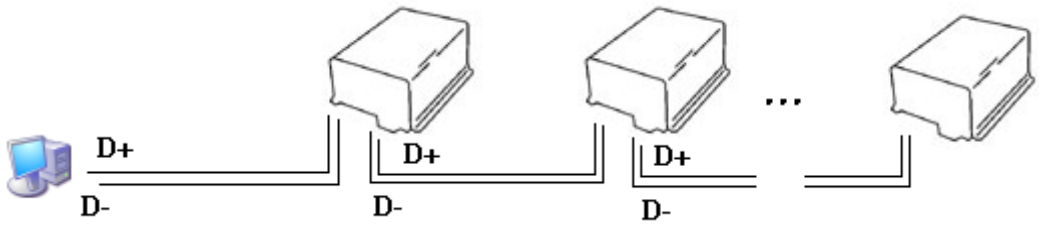

Figure 1

Modbus communication parameters:

Parameter	Description
Transfer mode	RTU mode
Communication mode	Half duplex
Baud rate	9600
Start bit	1
Data bit	8
Check bit	None
Stop bit	1

Technical terms:

Item	Description
Host	The one that initiates communication is called the host
Slave	The one that passive responses command called the slave
Broadcast address	0
Default address	1
U16	Unsigned integer of 16-bit
U32	Unsigned integer of 32-bit
S16	Signed integer of 16-bit
S32	Signed integer of 32-bit
CRC	Modbus CRC 16-bit, low byte in front
RO	Read only, only support 0x03 command
WO	Write only, only support 0x06 command
RW	Read and write, support 0x03、0x6、0x10 command

2 Register Definition

2.1 Device information

Index	Definition	Address	Register Number	Type	Comment	R/W	Unit
1	Device Model name	0x1A00	8	string	Ex.: PV 30KTL	RO	ASCII
2	Device Serial number	0x1A10	8	string	Ex.: 1234-123456789	RO	ASCII
3	Master firmware version	0x1A1C	3	string	Ex.: 123456	RO	ASCII
4	Slave firmware version	0x1A26	3	string	Ex.: 123456	RO	ASCII
5	MPPT Number	0x1A3B	1	U16	Ex.: 1=1 MPPT	RO	
6	Nominal Voltage	0x1A44	1	U16	Ex.: 2200 = 220V	RO	0.1V Note1
7	Nominal Frequency	0x1A45	1	U16	Ex.: 5000 = 50Hz	RO	0.01Hz Note2
8	Nominal Active Power (Low word)	0x1A46	1	U16	Ex.: 1500 = 1500W	RO	1W
9	Nominal Active Power (High word)	0x1A4E	1	U16	Ex.: 1 = (1*65536)W	RO	1W
10	Grid Phase Number	0x1A48	1	U16	1=single phase 2=split phase 3=three phases	RO	
11	EMS Firmware Version	0x1A60	3	string	Ex.: 123456	RO	ASCII
12	LCD firmware version	0x1A8E	3	string	Ex.: 123456	RO	ASCII

Note:

1. Nominal Voltage: Probably include of 200/220/230/240/380/440/460/480V power system.
2. Nominal Frequency: Include of 50/60Hz power system.

2.2 Inverter real time data

Index	Definition	Address	Register Number	Type	Comment	R/W	Unit
1	L1 Voltage	0x1001	1	U16		RO	0.1V
2	L1 Current	0x1002	1	U16		RO	0.01A
3	L1 Power	0x1003	2	S32		RO	0.1W
4	L1 Frequency	0x1005	1	U16		RO	0.01Hz
5	L2 Voltage	0x1006	1	U16		RO	0.1V
6	L2 Current	0x1007	1	U16		RO	0.01A
7	L2 Power	0x1008	2	S32		RO	0.1W
8	L2 Frequency	0x100A	1	U16		RO	0.01Hz
9	L3 Voltage	0x100B	1	U16		RO	0.1V
10	L3 Current	0x100C	1	U16		RO	0.01A
11	L3 Power	0x100D	2	S32		RO	0.1W
12	L3 Frequency	0x100F	1	U16		RO	0.01Hz
13	MPPT1 Voltage	0x1010	1	U16		RO	0.1V
14	MPPT1 Current	0x1011	1	S16		RO	0.01A
15	MPPT1 Power	0x1012	2	U32		RO	0.1W
16	MPPT2 Voltage	0x1014	1	U16		RO	0.1V
17	MPPT2 Current	0x1015	1	S16		RO	0.01A
18	MPPT2 Power	0x1016	2	U32		RO	0.1W
19	MPPT3 Voltage	0x1018	1	U16		RO	0.1V
20	MPPT3 Current	0x1019	1	S16		RO	0.01A
21	MPPT3 Power	0x101A	2	U32		RO	0.1W
22	Inner Temperature	0x101C	1	S16		RO	1℃
23	Inverter Mode	0x101D	1	U16	Inverter Mode Table	RO	
24	Error Code1	0x101E	1	U16	Error Code Table 1	RO	
25	Error Code2	0x101F	1	U16	Error Code Table 2	RO	
26	Error Code3	0x1020	1	U16	Error Code Table 3	RO	
27	Total Energy	0x1021	2	U32		RO	kWh
28	Total generation time	0x1023	2	U32		RO	Hour
29	Today Energy	0x1027	2	U32		RO	Wh
30	Fuse open data	0x1034	2	U16	Fuse Open Data Table	RO	
31	Grid total Active Power	0x1037	2	S32	(L1+L2+L3)	RO	0.1W
32	Grid total Reactive Power	0x1039	2	S32	(L1+L2+L3)	RO	0.1Var
33	PV Today Peak Power	0x103B	2	S32		RO	0.1W
34	Power Factor	0x103D	1	S16		RO	0.001

35	MPPT4 Voltage	0x103E	1	U16		RO	0.1V
36	MPPT4 Current	0x103F	1	S16		RO	0.01A
37	MPPT4 Power	0x1040	2	U32		RO	0.1W
38	PV total input power	0X1048	2	U32	(Add all MPPT)	RO	0.1W
39	Decimals of total energy	0x104C	1	U16	Relate to 0x1021, if total energy is 220.836kWh, then 0x104C is 836Wh	RO	Wh
40	MPPT5 Voltage	0x1080	1	U16		RO	0.1V
41	MPPT5 Current	0x1081	1	S16		RO	0.01A
42	MPPT5 Power	0x1082	2	U32		RO	0.1W
43	MPPT6 Voltage	0x1084	1	U16		RO	0.1V
44	MPPT6 Current	0x1085	1	S16		RO	0.01A
45	MPPT6 Power	0x1086	2	U32		RO	0.1W
46	MPPT7 Voltage	0x1088	1	U16		RO	0.1V
47	MPPT7 Current	0x1089	1	S16		RO	0.01A
48	MPPT7 Power	0x108A	2	U32		RO	0.1W
49	MPPT8 Voltage	0x108C	1	U16		RO	0.1V
50	MPPT8 Current	0x108D	1	S16		RO	0.01A
51	MPPT8 Power	0x108E	2	U32		RO	0.1W
52	MPPT9 Voltage	0x1090	1	U16		RO	0.1V
53	MPPT9 Current	0x1091	1	S16		RO	0.01A
54	MPPT9 Power	0x1092	2	U32		RO	0.1W
55	External fan speed duty percentage	0x120C	1	U16		RO	1%
56	DER-AVM digital meter connect status	0x1210	1	U16	0: disconnected 1: connected	RO	
57	Q (Reference Output reactive power)	0x1226	1	S16		RO	0.1kVAR
58	L1 watt of grid	0x1300	2	S32	+: Grid to inverter -: Inverter to grid	RO	0.1W
59	L2 watt of grid	0x1302	2	S32	+: Grid to inverter -: Inverter to grid	RO	0.1W
60	L3 watt of grid	0x1304	2	S32	+: Grid to inverter -: Inverter to grid	RO	0.1W
61	Accumulated energy of import	0x1306	2	U32	Grid to inverter	RO	10Wh
62	Accumulated energy of export	0x1308	2	U32	Inverter to grid	RO	10Wh
63	L1 watt of load	0x130A	2	U32		RO	0.1W

64	L2 watt of load	0x130C	2	U32		RO	0.1W
65	L3 watt of load	0x130E	2	U32		RO	0.1W
66	Accumulated energy of load	0x1310	2	U32		RO	10Wh
67	L1-N phase voltage of grid	0x131A	1	U16		RO	0.1V
68	L2-N phase voltage of grid	0x131B	1	U16		RO	0.1V
69	L3-N phase voltage of grid	0x131C	1	U16		RO	0.1V
70	L1 current of grid	0x131D	2	S32	+: Grid to inverter -: Inverter to grid	RO	0.01A
71	L2 current of grid	0x131F	2	S32	+: Grid to inverter -: Inverter to grid	RO	0.01A
72	L3 current of grid	0x1321	2	S32	+: Grid to inverter -: Inverter to grid	RO	0.01A
73	L1-N phase voltage of load	0x1323	1	U16		RO	0.1V
74	L2-N phase voltage of load	0x1324	1	U16		RO	0.1V
75	L3-N phase voltage of load	0x1325	1	U16		RO	0.1V
76	L1 current of load	0x1326	2	S32		RO	0.01A
77	L2 current of load	0x1328	2	S32		RO	0.01A
78	L3 current of load	0x132A	2	S32		RO	0.01A
79	Today import Energy	0x1332	2	U32		RO	10Wh
80	Today export Energy	0x1334	2	U32		RO	10Wh
81	Today load Energy	0x1336	2	U32		RO	10Wh
82	Frequency of grid	0x1338	1	U16		RO	0.01Hz
83	String1 input voltage	0x1050	1	U16		RO	0.1V
84	String1 input current	0x1051	1	S16		RO	0.01A
85	String2 input voltage	0x1052	1	U16		RO	0.1V
86	String2 input current	0x1053	1	S16		RO	0.01A
87	String3 input voltage	0x1054	1	U16		RO	0.1V
88	String3 input current	0x1055	1	S16		RO	0.01A
89	String4 input voltage	0x1056	1	U16		RO	0.1V
90	String4 input current	0x1057	1	S16		RO	0.01A
91	String5 input voltage	0x1058	1	U16		RO	0.1V
92	String5 input current	0x1059	1	S16		RO	0.01A
93	String6 input voltage	0x105A	1	U16		RO	0.1V
94	String6 input current	0x105B	1	S16		RO	0.01A
95	String7 input voltage	0x105C	1	U16		RO	0.1V
96	String7 input current	0x105D	1	S16		RO	0.01A
97	String8 input voltage	0x105E	1	U16		RO	0.1V
98	String8 input current	0x105F	1	S16		RO	0.01A

99	String9 input voltage	0x1060	1	U16		RO	0.1V
100	String9 input current	0x1061	1	S16		RO	0.01A
101	String10 input voltage	0x1062	1	U16		RO	0.1V
102	String10 input current	0x1063	1	S16		RO	0.01A
103	String11 input voltage	0x1064	1	U16		RO	0.1V
104	String11 input current	0x1065	1	S16		RO	0.01A
105	String12 input voltage	0x1066	1	U16		RO	0.1V
106	String12 input current	0x1067	1	S16		RO	0.01A
107	String13 input voltage	0x1068	1	U16		RO	0.1V
108	String13 input current	0x1069	1	S16		RO	0.01A
109	String14 input voltage	0x106A	1	U16		RO	0.1V
110	String14 input current	0x106B	1	S16		RO	0.01A
111	String15 input voltage	0x106C	1	U16		RO	0.1V
112	String15 input current	0x106D	1	S16		RO	0.01A
113	String16 input voltage	0x106E	1	U16		RO	0.1V
114	String16 input current	0x106F	1	S16		RO	0.01A
115	String17 input voltage	0x1070	1	U16		RO	0.1V
116	String17 input current	0x1071	1	S16		RO	0.01A
117	String18 input voltage	0x1072	1	U16		RO	0.1V
118	String18 input current	0x1073	1	S16		RO	0.01A
119	String19 input voltage	0x1074	1	U16		RO	0.1V
120	String19 input current	0x1075	1	S16		RO	0.01A
121	String20 input voltage	0x1076	1	U16		RO	0.1V
122	String20 input current	0x1077	1	S16		RO	0.01A
123	String21 input voltage	0x1078	1	U16		RO	0.1V
124	String21 input current	0x1079	1	S16		RO	0.01A
125	String22 input voltage	0x107A	1	U16		RO	0.1V
126	String22 input current	0x107B	1	S16		RO	0.01A
127	String23 input voltage	0x107C	1	U16		RO	0.1V
128	String23 input current	0x107D	1	S16		RO	0.01A
129	String24 input voltage	0x107E	1	U16		RO	0.1V
130	String24 input current	0x107F	1	S16		RO	0.01A
131	history log 0	0xB600	3	U16	NOTE 4	RO	
132	……	……	……	……		……	
133	history log 127	0xB77D	3	U16		RO	
134	Tracker 1 voltage	0x8000	64	U16		RO	0.1V
135	String 1-1 current	0x8040	64	S16		RO	0.01A

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136	String 1-2 current	0x8080	64	S16		RO	0.01A
137	String 1-3 current	0x80C0	64	S16		RO	0.01A
138	String 1-4 current	0x8100	64	S16		RO	0.01A
139	Tracker 2 voltage	0x8140	64	U16		RO	0.1V
140	String 2-1 current	0x8180	64	S16		RO	0.01A
141	String 2-2 current	0x81C0	64	S16		RO	0.01A
142	String 2-3 current	0x8200	64	S16		RO	0.01A
143	String 2-4 current	0x8240	64	S16		RO	0.01A
144	Tracker 3 voltage	0x8280	64	U16		RO	0.1V
145	String 3-1 current	0x82C0	64	S16		RO	0.01A
146	String 3-2 current	0x8300	64	S16		RO	0.01A
147	String 3-3 current	0x8340	64	S16		RO	0.01A
148	String 3-4 current	0x8380	64	S16		RO	0.01A
149	Tracker 4 voltage	0x83C0	64	U16		RO	0.1V
150	String 4-1 current	0x8400	64	S16		RO	0.01A
151	String 4-2 current	0x8440	64	S16		RO	0.01A
152	String 4-3 current	0x8480	64	S16		RO	0.01A
153	String 4-4 current	0x84C0	64	S16		RO	0.01A
154	Tracker 5 voltage	0x8500	64	U16		RO	0.1V
155	String 5-1 current	0x8540	64	S16		RO	0.01A
156	String 5-2 current	0x8580	64	S16		RO	0.01A
157	String 5-3 current	0x85C0	64	S16		RO	0.01A
158	String 5-4 current	0x8600	64	S16		RO	0.01A
159	Tracker 6 voltage	0x8640	64	U16		RO	0.1V
160	String 6-1 current	0x8680	64	S16		RO	0.01A
161	String 6-2 current	0x86C0	64	S16		RO	0.01A
162	String 6-3 current	0x8700	64	S16		RO	0.01A
163	String 6-4 current	0x8740	64	S16		RO	0.01A
164	Tracker 7 voltage	0x8780	64	U16		RO	0.1V
165	String 7-1 current	0x87C0	64	S16		RO	0.01A
166	String 7-2 current	0x8800	64	S16		RO	0.01A
167	String 7-3 current	0x8840	64	S16		RO	0.01A
168	String 7-4 current	0x8880	64	S16		RO	0.01A
169	Tracker 8 voltage	0x88C0	64	U16		RO	0.1V
170	String 8-1 current	0x8900	64	S16		RO	0.01A
171	String 8-2 current	0x8940	64	S16		RO	0.01A
172	String 8-3 current	0x8980	64	S16		RO	0.01A
173	String 8-4 current	0x89C0	64	S16		RO	0.01A
174	Tracker 9 voltage	0x8A00	64	U16		RO	0.1V

175	String 9-1 current	0x8A40	64	S16		RO	0.01A
176	String 9-2 current	0x8A80	64	S16		RO	0.01A
177	String 9-3 current	0x8AC0	64	S16		RO	0.01A
178	String 9-4 current	0x8B00	64	S16		RO	0.01A

Note:

1. L1/L2/L3 Voltage/Current/Power register address for all model even single phase or split phase model.
2. MPPT1~4 Voltage/Current/Power register address for all model.
3. MPPT5~9 Voltage/Current/Power register address depend on model.
4. Example：

   (1) Query history 1: 01 03 B6 00 00 03 23 83

   (2) Reply history 1: 01 03 06 06 B3 A4 97 00 05 76 16

   (3) Parsing：

   - Register 0xB600 = 0x06B3 = 0b000001 1010 110011

   Year=0b000001=1(add 2020); Month=0b1010=10; Second=0b110011=51

   - Register 0xB601 =0xA497= 0b10100 10010 010111

   Day=0b10100=20; Hour=0b10010=18; Minute=0b010111=23

   - Error Code= 0x0005=0b0 000000000000101

   flag=0 warning set; warning number=0b000000000000101=5

   Based on error code table, number 5 is means “PV over voltage"

   - So history 1：2021-10-20 18:23:51 “PV over voltage"

   (Format: YYYY-MM-DD hh:mm:ss history)

2.3 DEA-AVM real time data

Index	Definition	Address	Register Number	Type	Comment	R/W	Unit
1	L1 Current	0x03E8	2	S32		RO	0.1A
2	L2 Current	0x03EA	2	S32		RO	0.1A
3	L3 Current	0x03EC	2	S32		RO	0.1A
4	L1 Voltage	0x03EE	2	S32		RO	0.1V
5	L2 Voltage	0x03F0	2	S32		RO	0.1V
6	L3 Voltage	0x03F2	2	S32		RO	0.1V
7	Grid total Active Power	0x03F4	2	S32	(L1+L2+L3)	RO	0.1kW
8	Grid total Reactive Power	0x03F6	2	S32	(L1+L2+L3)	RO	Var
9	Power Factor	0x03F8	2	S32		RO	0.001
10	Frequency	0x03FA	2	S32		RO	0.1Hz
11	Status Flag 1	0x03FC	2	S32	Additional description	RO	

2.4 Inverter parameters for PV Inverter

Index	Definition	Address	Register Number	Type	Comment	R/W	Unit
1	DER-AVM mode	0x1211	1	U16	0: self control 1: DER-AVM control	RW	
2	RS485-2 Modbus address	0x1240	1	U16	[1, 255]	RW	
3	RS485-2 UART baud rate	0x1241	1	U16	[3, 6] 3: 9600 4: 14400 5: 19200 6: 38400	RW	
4	RS485-2 terminal resistance control	0x1242	1	U16	[0, 1] 0: open,1: close	RW	
5	RS485-3 Modbus address	0x1243	1	U16	[1, 255]	RW	
6	RS485-3 UART baud rate	0x1244	1	U16	[3, 6] 3: 9600 4: 14400 5: 19200 6: 38400	RW	
7	RS485-3 terminal resistance control	0x1245	1	U16	[0, 1] 0: open,1: close	RW	
8	Nominal voltage	0x1A44	1	U16		RW	0.1V
9	Date：Year	0x3000	1	U16	[2000, 2099]	RW	
10	Date：Month + Day	0x3001	1	U16	[1, 12] High Byte=Month [1, 31] Low Byte=Day	RW	
11	Date：Hour + Minute	0x3002	1	U16	[0, 23] High Byte =Hour [0, 59] Low Byte =Minute	RW	
12	Date：Second + 0	0x3003	1	U16	[0, 59] High Byte =Second Low Byte =0	RW	
13	Power derating percent by Modbus	0x3005	1	U16	[0, 110] ***Dynamic active power regulation.***	RW	
14	RS485-1 Modbus Address	0x303E	1	U16	[1, 247]	RW	

					Default 1		
15	RS485-1 baud rate	0x304C	1	U16	[3, 6] 3: 9600 4: 14400 5: 19200 6: 38400	RW	
16	WIFI STA SSID	0x3060	16	string		RW	ASCII
17	WIFI STA password	0x3070	16	string		RW	ASCII
18	Digital meter Modbus address	0x30B0	1	U16	[1, 255] Default 1	RW	
19	Digital meter Type	0x30B1	1	U16	[0, 10] 0: Unknown 1: Gavazzi/ EM340DINAV23XS1X 2: CHINT/DTSU666 3: Reserved 4: Lovato/DMG210 5: Reserved 6: Gavazzi/ET112 7: Reserved 8: CHINT/DDSU666 9: Gavazzi/ EM530DINAV53XS1X 10: Gavazzi/ EM540DINAV53XS1X	RW	
20	Power flow direction	0x30B2	1	U16	[0, 1] 0: Grid to inverter 1: Inverter to grid	RW	
21	Power limit function	0x30B3	1	U16	[0, 3] 0: Disable 1: Power limit by external device 2: Power limit by PVI external CT 3: Power limit by digital meter	RW	
22	Power limit CT ratio	0x30B4	1	U16	[0, 8] 0: Unknown 1: 1000 : 1	RW	

					2: 2000 : 1 3: 2500 : 1 4: 3000 : 1 5: 4000 : 1 6: 5000 : 1 7: 6000 : 1 8: 10000 : 1		
23	Meter location	0x30B5	1	U16	[0, 1] 0: CT on Grid (Meter on Grid) 1: CT on Load (Meter on Load)	RW	
24	Maximum feed in grid power	0x30B9	2	U32	[0, Pnom] Pnom : nominal power	RW	W
25	First connect start time	0x5000	1	U16	[10, 600]	RW	s
26	Reconnect time	0x5001	1	U16	[10, 900]	RW	s
27	Grid frequency high loss level 1 limit	0x5002	1	U16	50Hz: [5001, 5500] 60Hz: [6001, 6500]	RW	0.01Hz
28	Grid frequency low loss level 1 limit	0x5003	1	U16	50Hz: [4500, 4999] 60Hz: [5500, 5999]	RW	0.01Hz
29	Grid voltage high loss level 1 limit	0x5004	1	U16	[1, 1.5] * nominal voltage	RW	0.1V
30	Grid voltage low loss level 1 limit	0x5005	1	U16	[0.1, 1] * nominal voltage	RW	0.1V
31	Grid frequency high loss level 1 trip time	0x5006	1	U16	[0, 65535]	RW	1ms
32	Grid frequency low loss level 1 trip time	0x5007	1	U16	[0, 65535]	RW	1ms Note1
33	Grid voltage high loss level 1 trip time	0x5008	1	U16	[0, 65535]	RW	1ms Note1
34	Grid voltage low loss level 1 trip time	0x5009	1	U16	[0, 65535]	RW	1ms Note1
35	Grid frequency high loss level 2 limit	0x500A	1	U16	50Hz: [5001, 5500] 60Hz: [6001, 6500]	RW	0.01Hz
36	Grid frequency low loss level 2 limit	0x500B	1	U16	50Hz: [4500, 4999] 60Hz: [5500, 5999]	RW	0.01Hz

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37	Grid voltage high loss level 2 limit	0x500C	1	U16	[1, 1.5] * nominal voltage	RW	0.1V
38	Grid voltage low loss level 2 limit	0x500D	1	U16	[0.1, 1] * nominal voltage	RW	0.1V
39	Grid frequency high loss level 2 trip time	0x500E	1	U16	[0, 65535]	RW	1ms Note1
40	Grid frequency low loss level 2 trip time	0x500F	1	U16	[0, 65535]	RW	1ms Note1
41	Grid voltage high loss level 2 trip time	0x5010	1	U16	[0, 65535]	RW	1ms Note1
42	Grid voltage low loss level 2 trip time	0x5011	1	U16	[0, 65535]	RW	1ms Note1
43	Grid Frequency High Level 1 back	0x5012	1	U16	50Hz: [5001, 5500] 60Hz: [6001, 6500]	RW	0.01Hz
44	Grid Frequency Low Level 1 back	0x5013	1	U16	50Hz: [5001, 5500] 60Hz: [6001, 6500]	RW	0.01Hz
45	10 min average sustained voltage	0x5018	1	U16	[1, 1.5] * nominal voltage	RW	0.1V
46	Reconnect soft output power percent	0x5019	1	U16	[1,100]	RW	%/min
47	Over frequency power reduction droop	0x501A	1	U16	[20,120]	RW	0.1%/Hz
48	Insulation resistance active limit	0x501B	1	U16	[10,5000]	RW	1KΩ
49	Grid over voltage de-rating point	0x501E	1	U16	[1, 1.5] * nominal voltage	RW	0.1V
50	Grid frequency high Level 1 trip time (High word)	0x501F	1	U16	[0,15]	RW	ms
51	Grid frequency low Level 1 trip time (High word)	0x5020	1	U16	[0,15]	RW	ms
52	Grid over frequency de-rating start point	0x5021	1	U16	50Hz: [5001, 5500] 60Hz: [6001, 6500]	RW	0.01Hz
53	Grid over frequency de-rating end point	0x5022	1	U16	50Hz: [5001, 5500] 60Hz: [6001, 6500]	RW	0.01Hz
54	Grid Voltage High Level 1 trip time (High word)	0x5025	1	U16	[0,9]	RW	ms
55	Grid Voltage Low Level 1 trip time (High word)	0x5026	1	U16	[0,9]	RW	ms
56	Grid Voltage High Level 1	0x5027	1	U16	[nominal voltage + 1,	RW	0.1V

	back				nominal voltage * 1.5]		
57	Grid Voltage Low Level 1 back	0x5028	1	U16	[0, nominal voltage - 1]	RW	0.1V
58	First connect soft start output power percent	0x5029	1	U16	[1,100]	RW	%
59	Output reactive power mode	0x5030	1	U16	[0,9] 0: Pure active power 1: cosφ = const. relation with [0x5031] 2: Q = const. relation with [0x5032] | [0x5114] 3: cosφ(P), relation with [0x5034~0x503B], [0x5045,0x5046 CEI-021 only] 4: Q(U), relation with [0x503C~0x5044], [0x5047,0x5048 CEI-021 & PEA only] 5: AUTO P(U) 6: Watt(U) AS4777, Pure active power, relation with [0x5049~0x5050] 7: Q(U) AS4777, relation with [0x5051~0x5058] 8: Q = const. (P priority), relation with [0x5114] 9: Q(P) Curve, relation with [0x505A~0x5061]	RW	
60	Power factor setting	0x5031	1	S16	[-1000, -800], [800, 1000]	RW	0.001
61	Reactive control response time	0x5033	1	U16	[0,120]	RW	s
62	Q(U) curve node1 percent(U/Un)	0x503C	1	U16	[80,120]	RW	%
63	Q(U) curve node2 percent(U/Un)	0x503D	1	U16	[80,120]	RW	%
64	Q(U) curve node3 percent(U/Un)	0x503E	1	U16	[80,120]	RW	%

65	Q(U) curve node4 percent(U/Un)	0x503F	1	U16	[80,120]	RW	%
66	Q(U) curve node1 value setting (Q/S)	0x5040	1	S16	[-1100,1100]	RW	0.1%
67	Q(U) curve node2 value setting (Q/S)	0x5041	1	S16	[-1100,1100]	RW	0.1%
68	Q(U) curve node3 value setting (Q/S)	0x5042	1	S16	[-1100,1100]	RW	0.1%
69	Q(U) curve node4 value setting (Q/S)	0x5043	1	S16	[-1100,1100]	RW	0.1%
70	Volt-Watt node1 voltage for output power derating	0x504B	1	U16	[0.8, 1.5] * nominal voltage	RW	0.1V
71	Volt-Watt node2 voltage for output power derating	0x504C	1	U16	[0.8, 1.5] * nominal voltage	RW	0.1V
72	Volt-Watt node1 output power percent	0x504F	1	U16	[0,110]	RW	%
73	Volt-Watt node2 output power percent	0x5050	1	U16	[0,110]	RW	%
74	Q(P) curve node 1 percentage setting (P/Pn)	0x505A	1	U16	[0,100]	RW	%
75	Q(P) curve node 2 percentage setting (P/Pn)	0x505B	1	U16	[0,100]	RW	%
76	Q(P) curve node 3 percentage setting (P/Pn)	0x505C	1	U16	[0,100]	RW	%
77	Q(P) curve node 4 percentage setting (P/Pn)	0x505D	1	U16	[0,100]	RW	%
78	Q(P) curve node 1 value setting (Q/Pn)	0x505E	1	S16	[-600,600]	RW	0.10%
79	Q(P) curve node 2 value setting (Q/Pn)	0x505F	1	S16	[-600,600]	RW	0.10%
80	Q(P) curve node 3 value setting (Q/Pn)	0x5060	1	S16	[-600,600]	RW	0.10%
81	Q(P) curve node 4 value setting (Q/Pn)	0x5061	1	S16	[-600,600]	RW	0.10%
82	LVRT Triggering voltage	0x5063	1	U16	[0.1, 0.9] * nominal voltage	RW	0.1V
83	HVRT Triggering voltage	0x5064	1	U16	[1.1, 1.5] * nominal voltage	RW	0.1V
84	FRT Mode	0x5065	1	U16	[0, 1]	RW	

					0: Zero current mode, 1: full grid support mode		
85	FRT Zero current mode triggering threshold	0x5066	1	U16	[5, 150]	RW	%
86	FRT grid voltage jump triggering threshold	0x5067	1	U16	[5, 150]	RW	%
87	Grid frequency high Level 2 trip time (High word)	0x506C	1	U16	[0, 15]	RW	ms
88	Grid frequency low Level 2 trip time (High word)	0x506D	1	U16	[0, 15]	RW	ms
89	Grid frequency high level 3 limit	0x5070	1	U16	50Hz: [5001, 5500] 60Hz: [6001, 6500]	RW	0.01Hz
90	Grid frequency low level 3 limit	0x5071	1	U16	50Hz: [4500, 4999] 60Hz: [5500, 5999]	RW	0.01Hz
91	Grid voltage high level 3 limit	0x5072	1	U16	[1, 1.5] * nominal voltage	RW	0.1V
92	Grid voltage low level 3 limit	0x5073	1	U16	[0.1, 1] * nominal voltage	RW	0.1V
93	Grid Frequency High Level 3 trip time	0x5074	1	U16	[0, 65535]	RW	ms
94	Grid Frequency Low Level 3 trip time	0x5075	1	U16	[0, 65535]	RW	ms
95	Grid Voltage High Level 3 trip time	0x5076	1	U16	[0, 65535]	RW	ms
96	Grid Voltage Low Level 3 trip time	0x5077	1	U16	[0, 65535]	RW	ms
97	Over frequency derating reference power	0x507F	1	U16	[0, 1] 0: base on rating power, 1: base on current power	RW	
98	Grid under frequency rise power start point	0x5081	1	U16	50Hz: [4500, 4999] 60Hz: [5500, 5999]	RW	0.01Hz
99	Grid under frequency rise power stop point	0x5082	1	U16	50Hz: [4500, 4999] 60Hz: [5500, 5999]	RW	0.01Hz
100	Under frequency rise power droop	0x5083	1	U16	[20, 120]	RW	0.001
101	Under frequency rise power reference	0x5084	1	U16	[0, 1] 0: base on rating power, 1: base on current power	RW	
102	Regulation code	0x5101	1	U16	Regulation Code Table	RW	

103	Derating Watt Percent	0x5104	1	U16	[0, 110] Percent of nominal power		%
104	Mppt shadow manage	0x5106	1	U16	[0, 1] 0: disable,1: enable	RW	
105	Mppt shadow manage interval time	0x5107	1	U16	[30, 7200]	RW	s
106	Islanding detection	0x510E	1	U16	[0, 1] 0: disable,1: enable	RW	
107	Ground current limit	0x5110	1	U16	[1, S*0.01]	RW	mA
108	Low voltage through detection	0x5112	1	U16	[0, 1] 0: disable,1: enable	RW	Note2
109	Reactive Power Percent	0x5114	1	S16	[-60, 60] Percent of nominal power ***Do not use for dynamic regulation.***	RW	%
110	RS485-1 terminal resistance control	0x5115	1	U16	[0, 1] 0: open,1: close	RW	Note2
111	Insulation Resistor Detection	0x5117	1	U16	[0, 1] 0: disable,1: enable	RW	
112	Ground Current Detection	0x5118	1	U16	[0, 1] 0: disable,1: enable	RW	
113	Grid high voltage load de-rating function	0x511D	1	U16	[0, 1] 0: disable,1: enable Relation with [0x501E]	RW	
114	Reactive Power Percent	0x5121	1	S16	[-11000, 11000]	RW	0.01%
115	Fuse open Check Mask	0x513D	2	U32	Fuse open Check Mask Table	RW	
116	Inverter control	0x6001	1	U16	[0, 1] 0: power on,1: shut down	RW	
117	EEPROM default	0x6006	1	U16	[0, 1] 0: EEPROM setting NOT to default	RW	

					1: EEPROM setting to default		
118	Clear the EEPROM history energy record	0x6009	1	U16	[0, 1] 0: NOT clear the EEPROM history energy record 1: Clear the EEPROM history energy record	RW	
119	I-V curve scan	0x600D	1	U16	Write: 0x0000: Non-active 0x0001: Active Read: 0x0000: Idle or had been read 0x0001: Running 0x0002: Finished	RW	
120	P.F (cosφ)	0x600F	1	S16	[-1000, -800], [800, 1000] (+) Positive: over-excited (lag, inductive) (-) Negative: under-excited (lead, capacitive) ***Dynamic power factor regulation.***	RW	0.001
121	Reactive Power Percent	0x6010	1	S16	[0, 65535] (+) Positive: over-excited (lag, inductive) (-) Negative: under-excited (lead, capacitive) ***Dynamic reactive power regulation.***	RW	0.01%

Note:

1. The unit of CN (NB/T 32004) regulation code (0x5101= 0x0E) is 10ms. The unit of other regulation codes is 1ms.
2. Only support for 3 phase model, means 0x1A48 = 3.
3. 0x5000/0x5100 write parameter to EEPROM, write more frequently will reduce the EEPROM life, so forbid send 0x5000/0x5100 group command to do active and reactive power control.
4. Suggestion send 0x3005 to dynamic control active power percent.

5. Suggestion send 0x600F/0x6010 to dynamic control the power factor or reactive power adjust function.
6. If setting range is [0, 65535], it means the setting range probably be limited by PV inverter nominal power, if setting value over nominal power will get the illegal code.
7. If setting range is [1, 1.5] * nominal voltage, the nominal voltage parameter needs reference the 0x1A44 parameter, for example, if 0x1A44 response value is 2200, it’ s means the setting range is [2200, 3300], if 0x1A44 response value is 4800, it’ s means the setting range is [4800, 7200]
8. If setting range is [0.1, 1] * nominal voltage, the nominal voltage parameter needs reference the 0x1A44 parameter, for example, if 0x1A44 response value is 2200, it’ s means the setting range is [220, 2200], if 0x1A44 response value is 4800, it’ s means the setting range is [480, 4800]
9. If setting range is [0, Pnom], the Pnom parameter need reference the 0x1A46 & 0x1A4E parameter, for example, if 0x1A46 response value is 10000 and 0x1A4E response value is 0, it’ s means the setting range is [0, 10000]

2.5 DER-AVM parameters for PV Inverter

Index	Definition	Address	Register Number	Type	Comment	R/W	Unit
1	Power factor setting	0x07D0	1	S16	[-1000, -800], [800, 1000] Default 1000	RW	0.001
2	Action mode	0x07D1	1	U16	0: self-control 2: DER-AVM control 5: Q(V)control Default 2	RW	
3	Reactive power percent setting	0x07D2	1	S16	[-484, 484] Default 0	RW	0.1%
4	Active power percent setting	0x07D3	1	U16	[0, 1100] Default 1000	RW	0.1%
5	Inverter off	0x0834	1	U16	0: Inverter on 1: Inverter off Default 0	RW	
6	Clear PV insulation warning	0x0835	1	U16	0: Non-active 1: Clear PV insulation abnormal warning	RW	

3 Modbus Protocol Command

3.1 Function code list

Index	Function code	Description
1	0x03	Read register address
2	0x06	Write a single register address
3	0x10	Write multiple register address
4	0x7E	REMS read register address

3.2 Read Register (0x03)

(1) Host query command format:

Parameter	Length (Byte)	Description
Slave address	1	1~247
Function code	1	0x03
Register start address	2	0x0000~0xFFFF
Register number	2	1~124
CRC code	2	

(2) Slave normal respond:

Parameter	Length (Byte)	Description
Slave address	1	1~247
Function code	1	0x03
Byte number	1	Register Number*2
Register value	2 ~ 248	
CRC code	2	

(3) Slave abnormal respond:

Parameter	Length (Byte)	Description
Slave address	1	1~247
Function code	1	0x83
Illegal code	1	Refer "Illegal CodeTable"
CRC code	2	

(4) Example:

Host query command : 01 03 10 01 00 01 D1 0A

Slave normal respond : 01 03 02 08 FC BF C5

Slave abnormal respond: 01 83 02 C0 F1

3.3 Write a single Register (0x06)

(1) Host query command format:

Parameter	Length (Byte)	Description
Slave address	1	1~247
Function code	1	0x06
Register address	2	0x0000~0xFFFF
Register value	2	0x0000~0xFFFF
CRC code	2	

(2) Slave normal respond:

Parameter	Length (Byte)	Description
Slave address	1	1~247
Function code	1	0x06
Register address	2	0x0000~0xFFFF
Register value	2	0x0000~0xFFFF
CRC code	2	

(3) Slave abnormal respond:

Parameter	Length (Byte)	Description
Slave address	1	1~247
Function code	1	0x86
Illegal code	1	Refer "Illegal CodeTable"
CRC code	2	

(4) Example:

Host query command : 01 06 51 01 00 01 09 36

Slave normal respond : 01 06 51 01 00 01 09 36

Slave abnormal respond: 01 86 04 43 A3

3.4Write multiple Registers (0x10)

(1) Host query command format:

Parameter	Length (Byte)	Description
Slave address	1	1~247
Function code	1	0x10
Register start address	2	0x0000~0xFFFF
Register number	2	1~122
Byte number	1	Register Number * 2
Register value	2 ~ 244	
CRC code	2	

(2) Slave normal respond:

Parameter	Length (Byte)	Description
Slave address	1	1~247
Function code	1	0x10
Register start address	2	0x0000~0xFFFF
Register number	2	1~122
CRC code	2	

(3) Slave abnormal respond:

Parameter	Length (Byte)	Description
Slave address	1	1~247
Function code	1	0x90
Illegal code	1	Refer "Illegal CodeTable"
CRC code	2	

(4) Example:

Host query command : 01 10 30 00 00 04 08 07 E1 01 01 00 00 00 00 7B 73

Slave normal respond : 01 10 30 00 00 04 CE CA

Slave abnormal respond: 01 90 02 CD C1

3.5REMS read register address(0x7E)

(1) Host query command format:

Parameter	Length (Byte)	Description
Function code	1	0x7E

Slave address	1	1~247
Command	1	0x07
CRC code	2	1~122

(2) Slave normal respond:

Parameter	Length (Byte)	Description
Function code	1	0x7E
Slave address	1	1~247
Command	1	0x08
Byte number	2	0x26
Register value	38	
CRC code	2	

(3) Example:

Host query command : 7E 01 07 51 8A

Slave normal respond : 7E 01 08 00 26 00 6E 00 0A 00 00 22 DE 01 9B 01 97 01 9A 00 0C 00 0C 00 0C 00 00 1F A8 03 E1 01 F4 00 00 00 00 48 2F C8 38 00 00 56 6C

4 Appendix

4.1 Error Code Table1 (0x101E)

Bit	No	Description
0	16	Inverter over dc-bias current
1	17	Inverter relay abnormal
2	18	Remote off
3	19	Inverter over temperature
4	20	GFCI abnormal
5	21	PV string reverse
6	22	System type error
7	23	Fan abnormal
8	24	Dc-link unbalance or under voltage
9	25	Dc-link over voltage
10	26	Internal communication error
11	27	Software incompatibility
12	28	Internal storage error
13	29	Data inconsistency
14	30	Inverter abnormal
15	31	Boost abnormal

4.2 Error Code Table2 (0x101F)

Bit	No	Description
0	0	Grid over voltage
1	1	Grid under voltage
2	2	Grid absent
3	3	Grid over frequency
4	4	Grid under frequency
5	5	PV over voltage
6	6	PV insulation abnormal
7	7	Leakage current abnormal
8	8	Inverter in power limit state
9	9	Internal power supply abnormal
10	10	PV string abnormal
11	11	PV under voltage
12	12	PV irradiation weak
13	13	Grid abnormal

14	14	Arc fault detection
15	15	AC moving average voltage high

4.3 Error Code Table3 (0x1020) ↺

Bit	No	Description
0	32	Reserved
1	33	Logger/E-Display EEPROM fail
2	34	Reserved
3	35	Single tracker detect warning (PID device abnormal)
4	36	AFCI lost
5	37	Data logger lost
6	38	Meter lost
7	39	Inverter lost
8	40	Grid N abnormal
9	41	Surge Protection Devices (SPD) defective
10	42	Parallel ID warning
11	43	Parallel SYN signal warning
12	44	Parallel BAT abnormal
13	45	Parallel GRID abnormal
14	46	Generator voltage abnormal
15	47	Reserved

4.4 Regulation code Table (0x5101) ↺

No.	Regulation code
0	0xFFFF : UNKNOWN
1	0x0001 : AU (澳洲 AS/NZS 4777.2 Region B)
2	0x0002 : DE (德國 VDE 0126-1-1/A1)
3	0x0003 : TW (台灣 TW GRID)
4	0x0004 : DE (德國 VDE-AR-N 4105:2011)
5	0x0005 : JP (日本 JETGR0002-1-2.0)
6	0x0006 : IT (義大利 CEI 0-21)
7	0x0007 : SE (瑞典 EIFS-2)
8	0x0008 : UK (英國 G98)
9	0x0009 : UL (美國 UL)
10	0x000A : TH(泰國 PEA)
11	0x000B : SE (瑞典 SWEDEN GRID) 2007

12	0x000C : NL (荷蘭 EN50549-1 2019)
13	0x000D : TH (泰國 MEA)
14	0x000E : CN (中國 NB/T 32004)
15	0x000F : IN (印度 IEC61727)
16	0x0010 : AU (澳洲 AS/NZS 4777.2 Region A)
17	0x0011 : NZ (澳洲 AS/NZS 4777.2 New Zealand)
18	0x0012 : US (美洲 IEEE1547)
19	0x0013 : UA (烏克蘭 VDE-AR-N 4105.)
20	0x0014 : TW (台灣 CNS 15382:2018)
21	0x0015 : DE (德國 VDE0126-1-1/08.13)
22	0x0016 : BR (巴西 ABNT NBR16149 2013)
23	0x0017 : CZ (捷克 Czech EN 50160)
24	0x0018 : UK (英國 G99 Type A)
25	0x0019 : PT (葡萄牙 EN50549-1)
26	0x001A : ES (西班牙 Spain RD1699 / RD413)
27	0x001B : ES (西班牙 Spain RD661 / PO12.2)
28	0x001C : IT (義大利 CEI 0-21 ACEA)
29	0x001D : DE (德國 VDE-AR-N 4105:2018)
30	0x001E : BR (巴西 IEC61727)
31	0x001F : EU (歐洲 EN50549-1:2019)
32	0x0020 : BE (比利时 C10/11 LV)
33	0x0021 : PL (波蘭 EN50438 2013)
34	0x0022 : DE (德國 VDE4110:2018)
35	0x0023 : EU (歐洲 EN50549-2:2019)
36	0x0024 : IT (義大利 CEI 0-16)
37	0x0025 : ZA (南非 NRS097-2-1)
38	0x0026 : GR (希臘 VDE0126_Continent)
39	0x0027 : GR (希臘 VDE0126_Islands)
40	0x0028 : JO (約旦 IRR-DCC MV 2015)
41	0x0029 : ES (西班牙 Spain UNE206007 / UNE206006)
42	0x002A : AU (澳洲 AS/NZS 4777.2 Region C)
43	0x002B : UK (英國 G99 Type B)
44	0x002C : JO (約旦 IRR-TIC 2015)
45	0x002D : CZ (捷克 Czech EN50549-1)
46	0x002E : AT (奧地利 TOR TypeA)
47	0x002F : CY (塞浦路斯 Cyprus 2019)
48	0x0030 : MX (墨西哥 IEEE1547 : 2014)

49	0x0031 : CL (智利 Chile BT)
50	0x0032 : FR (法國 UTE C15-712-1)
51	0x0033 : CH (瑞士 NA/EEA A2)
52	0x0034 : CH (瑞士 NA/EEA B2)
53	0x0035 : FR (法國 UTE C15-712-1 Island 50Hz)
54	0x0036 : FR (法國 UTE C15-712-1 Island 60Hz)
55	0x0037 : FI (芬蘭 VJV Type A)
56	0x0038 : FI (芬蘭 VJV Type B)
57	0x0039 : NO (挪威 EN50549-1)
58	0x003A : PL (波蘭 NC RFG Type A)
59	0x003B : PL (波蘭 NC RFG Type B)
60	0x003C : BE (比利时 C10/11 MV)
61	0x003D : AT (奥地利 TOR Type B)
62	0x003E : IL (以色列 IS 4777)
63	0x003F : DK1 (西丹麥 Type A & B LV)
64	0x0040 : DK1 (西丹麥 Type B MV)
65	0x0041 : DK2 (東丹麥 Type A & B LV)
66	0x0042 : DK2 (東丹麥 Type B MV)
67	0x0043 : CL (智利 Chile MT HD)
68	0x0044 : CL (智利 Chile MT LD)
69	0x0045 : ES (西班牙 NTs-A/UNE217002/RD647)
70	0x0046 : ES (西班牙 NTs-B/UNE217002/RD647)
71	0x0047 : AU (澳洲 AS/NZS 4777.2 Vector)
72	0x0048 : IRL (愛爾蘭 EN50549-1： Ireland)

4.5 Fuse Open Data Table (0x1034~0x1035)

Bit	Description
1	PV1 fuse open data, [0,1] 0:normal , 1: Fuse open
2	PV2 fuse open data, [0,1] 0:normal , 1: Fuse open
3	PV3 fuse open data, [0,1] 0:normal , 1: Fuse open
4	PV4 fuse open data, [0,1] 0:normal , 1: Fuse open
5	PV5 fuse open data, [0,1] 0:normal , 1: Fuse open

6	PV6 fuse open data, [0,1] 0:normal , 1: Fuse open
7	PV7 fuse open data, [0,1] 0:normal , 1: Fuse open
8	PV8 fuse open data, [0,1] 0:normal , 1: Fuse open

4.6 Inverter Mode Table (0x101D) ↺

No.	Description
0x00	Initial mode
0x01	Standby mode
0x03	On-Grid mode
0x04	Off-Grid mode
0x05	Fault mode
0x09	Shutdown mode

4.7 Illegal code Table ↺

Code	Description
01	Indicate that function code is not expected code of 0x03, 0x06, 0x10
02	Indicate that read or write register number is too large
03	Indicate that read or write register address is out of range
04	Indicate that read or write register value is out of limit or the register is forbidden to write

4.8 Fuse open Check Mask Table (0x513D~0x513E)

Bit	Description
1	PV1 fuse open check mask, [0,1] 0:Enable , 1: Disable
2	PV2 fuse open check mask, [0,1] 0:Enable , 1: Disable
3	PV3 fuse open check mask, [0,1] 0:Enable , 1: Disable
4	PV4 fuse open check mask, [0,1] 0:Enable , 1: Disable
5	PV5 fuse open check mask, [0,1]

	0:Enable , 1: Disable
6	PV6 fuse open check mask, [0,1] 0:Enable , 1: Disable
7	PV7 fuse open check mask, [0,1] 0:Enable , 1: Disable
8	PV8 fuse open check mask, [0,1] 0:Enable , 1: Disable

4.9Reference document

APD PV inverter Modbus communication protocol V4.25.xlsx

4.10 Additional description (0x03FC~03FD)

Bit	Description
1	Inverter action state, [0,1] 0:Run , 1: Fail
2	Inverter CB action state, [0,1] 0: Run, 1: Fail
3	Run state, [0,1] 0: DER-AVM control, 1: Self control
4	Active power control(communication), [0,1] In DER-AVM, the data is switched to 1 after a successful acceptance and to 0 after the next read
5	Inverter action control(communication), [0,1] In DER-AVM, the data is switched to 1 after a successful acceptance and to 0 after the next read

4.11 REMS register value

No.	Description	Length (Byte)	Unit
1	PV average voltage	2	V
2	PV total current	2	A
3	PV totally input power	4	W
4	L1 Voltage	2	V
5	L2 Voltage	2	V
6	L3 Voltage	2	V
7	L1 Current	2	A
8	L2 Current	2	A

9	L3 Current	2	A
10	Totally active power	4	W
11	Power Factor	2	0.1%
12	Frequency	2	0.1Hz
13	Total Energy	8	Wh
14	REMS Error Code Table	2	

4.12 REMS Error Code Table

Bit	Description
0	Inverter action state, [0,1] 0: inverter running, 1: inverter stop
1	Solar battery over voltage, [0,1] 0: normal, 1: over voltage
2	Solar battery under voltage, [0,1] 0: normal, 1: under voltage
3	Solar battery over current, [0,1] 0: normal, 1: over current
4	Inverter IGBT error, [0,1] 0: normal, 1: error
5	Inverter over temperature, [0,1] 0: normal, 1: over temperature
6	Grid over voltage, [0,1] 0: normal, 1: over voltage
7	Grid under voltage, [0,1] 0: normal, 1: under voltage
8	Grid over current, [0,1] 0: normal, 1: over current
9	Grid over frequency, [0,1] 0: normal, 1: over frequency
10	Grid under frequency, [0,1] 0: normal, 1: under frequency
11	Self control(outage), [0,1] 0: normal, 1: outage
12	Leakage current abnormal, [0,1] 0: normal, 1: abnormal